1. INTRODUCTION

Internships are a vital component of professional learning as they bridge the gap between the theoretical knowledge gained in academics and the practical skills required in the industry. They provide students with the opportunity to apply classroom learning in real-world situations, gain professional exposure, and understand the organizational work culture in a dynamic environment. As part of my academic program, I had the privilege to undergo a Software Developer Internship at Incisive Design and Business Solutions Private Limited, Coimbatore, Tamil Nadu, from 21st May 2025 to 21st June 2025. This internship was an important milestone in my journey as a budding software developer as it offered me a platform to acquire hands-on experience in frontend and backend development, work with modern technologies, and gain exposure to the complete life cycle of developing professional web applications. It not only strengthened my technical knowledge but also improved my soft skills such as communication, adaptability, time management, and teamwork. During the internship, I was able to contribute to real-time project tasks, which challenged my problem-solving abilities and helped me understand the practical implementation of concepts.

Incisive Design and Business Solutions Private Limited is a reputed IT services company that focuses on delivering customized business solutions tailored to client needs. The company emphasizes building modern, responsive, and user-friendly applications while offering services in frontend and backend development, database management, mobile app development, and cloud-based deployments. It promotes innovation by encouraging interns and employees to explore and work with the latest technologies such as Next.js, React.js, Tailwind CSS, Node.js, Express.js, MySQL, and Flutter. This provided me with the right environment to work on projects involving cutting-edge tools and frameworks that are highly relevant in the software industry today. My key objectives during the internship included gaining exposure to full-stack development processes, creating responsive and user-friendly web applications, building visually appealing user

interfaces, designing backend logic and RESTful APIs, working with MySQL databases, integrating frontend and backend modules, developing a mock mobile application using Flutter, and understanding processes like debugging, testing, deployment, and optimization.

This internship proved to be a highly significant and enriching learning experience as it allowed me to apply academic knowledge to practical projects, deepening my understanding of software engineering concepts. I acquired the ability to critically analyze problems and design efficient solutions, which is essential for a successful software development career. The experience of handling both frontend and backend responsibilities boosted my confidence in full-stack development, while close collaboration with senior developers taught me how to work in a team-oriented environment and manage tasks effectively within deadlines. Overall, the internship at Incisive Design and Business Solutions Pvt. Ltd. provided me with a perfect blend of technical learning, industry exposure, and professional growth, serving as a strong stepping stone for building my career in software development and full-stack engineering.

2. COMPANY PROFILE

Name: Incisive Design and Business Solutions Private Limited

Location: Coimbatore, Tamil Nadu, India

2.1 Overview

Incisive Design and Business Solutions Pvt. Ltd. is an IT services & solutions provider that specializes in building software applications for clients, with a strong orientation toward modern web and mobile technologies. Their work encompasses both frontend and backend systems, with attention to database management, user interface design, responsiveness, and deployment.

2.2 Vision / Mission

- delivering **user-friendly, responsive, and efficient software products**;
- staying up to date with current frameworks and technologies;
- providing a learning and growth environment for its team and interns.

2.3 Core Technologies & Expertise

- **Frontend frameworks**: Next.js, React.js
- **Styling / UI**: Tailwind CSS
- **Backend development**: Node.js, Express.js
- **Database**: MySQL
- **Mobile / cross-platform UI**: Flutter (for mock app development)
- **Deployment / cloud readiness** and performance optimization (e.g. optimizations, SEO)

2.4 Key Services & Offerings

- Full-stack web application development
- RESTful API creation and backend logic implementation

- User interface and UX design for responsive web and mobile
- Database schema design, queries, data storage & retrieval
- Integration of frontend and backend modules for smooth functionality
- Application debugging, testing, deployment & performance optimization

2.5 Organizational Culture & Strengths

- Emphasis on practical learning: Interns are given responsibilities across the stack, not just superficial tasks.
- Use of modern & in-demand technologies, which keeps the company relevant and enhances employee/intern growth.
- Work includes both visual / UI/UX components as well as backend logic, which means balanced exposure.
- Value placed on soft skills—teamwork, adaptability, effective communication, attention to detail.

2.6 Opportunities

- You gain exposure to both frontend and backend work, which helps build well-rounded skills.
- Real-world applications (not just toy problems) offer good understanding of deployment, performance concerns, and end-to-end workflows.
- Working with technologies that are highly relevant in current web development markets.
- Develop soft skills (teamwork, code review, deadlines, debugging) in a professional setting.

3. SYSTEM REQUIREMENTS

The internship projects required a set of hardware and software resources to ensure smooth development, testing, and deployment. The system specifications are categorized into hardware requirements and software requirements**.**

3.1 Hardware Requirements

- **Processor**: Intel Core i5 / AMD Ryzen 5 or higher
- **Clock Speed**: Minimum 2.4 GHz
- **RAM**: 8 GB (16 GB recommended for smooth multitasking and running IDEs/emulators)
- **Hard Disk**: 500 GB (1 TB recommended, preferably SSD for faster operations)
- **Display**: 1366 × 768 resolution (1920 × 1080 Full HD recommended)
- **Graphics**: Integrated Graphics (Dedicated GPU optional for Flutter mobile app development)
- **Peripherals**: Standard keyboard, mouse, and webcam
- **Network**: Stable internet connection (10 Mbps or higher) for API calls, version control, and deployments

3.2 Software Requirements

- **Operating System**: Windows 10 / 11 (64-bit)
- **Programming Languages**:
    - JavaScript (ES6+)
    - SQL
    - Dart (for Flutter development)
- **Frameworks & Libraries**:
    - **Frontend**: React.js, Next.js, Tailwind CSS
    - **Backend**: Node.js, Express.js

    - **Mobile**: Flutter SDK

- **Database**: MySQL Community Server 8.0 or higher
- **Tools & IDEs**:
    - Visual Studio Code (for React, Next.js, Node.js)
    - Android Studio (for Flutter mobile app)
    - MySQL Workbench (for database management)
- **Version Control**: Git and GitHub
- **Browsers**: Google Chrome / Mozilla Firefox (latest version)
- **Package Managers**: npm (Node Package Manager)
- **Other Utilities**:
    - Postman (for API testing)
    - Docker (optional, for containerized deployments)

4. APPLICATION USAGE

During my internship, I worked on two applications that focused on both web and mobile platforms. These applications were designed to provide hands-on exposure to real-world development and demonstrate the integration of different technologies for solving practical problems.

4.1 Flutter Mock Mobile Application

The Flutter mock app was developed as a prototype to demonstrate mobile-friendly user interfaces for the company's platform. It highlighted the importance of cross-platform mobile development, where a single codebase can be used to deploy applications on both Android and iOS.

- The app provides a mock interface simulating key modules of the company's official application.
- Designed with Flutter's widget-based framework, it showcased screens such as login, user dashboard, and basic navigation**.**
- The mock app helped visualize how real-time business solutions would appear on mobile devices.
- It allowed testing of UI layouts, responsiveness, and user interactions without requiring full backend integration.
- By using the app, the team could gather feedback from stakeholders on interface design and make improvements before full development.

4.2 Full-Stack Web Application (Next.js, React.js, Tailwind CSS, Node.js, Express.js, MySQL)

The full-stack web application was the major project of my internship. It was designed to demonstrate the complete frontend–backend–database integration using modern technologies.

- The application was used to manage user data and perform CRUD (Create, Read, Update, Delete) operations efficiently.
- The frontend (Next.js + React.js + Tailwind CSS) provided a responsive and user-friendly interface with consistent design and smooth navigation.
- Backend APIs (Node.js + Express.js) handled business logic, authentication, and data processing.
- The MySQL database was used to store structured information such as user details, application data, and records for easy retrieval.
- The application allowed real-time synchronization between frontend and backend, ensuring smooth data flow.
- Features like form submissions, data display in tables, login/registration, and admin panel controls made it practical and close to real-world use cases.
- The project helped demonstrate industry-ready software practices, such as RESTful API design, error handling, and performance optimization.

4.3 Overall Usage and Impact

Both applications served as learning platforms as well as proof-of-concept solutions for the company. The Flutter mock app allowed visualization of a mobile interface for clients and internal discussions, while the full-stack web application demonstrated the ability to build complete, responsive, and scalable solutions. Together, these projects showcased how modern technologies can be used to deliver seamless user experiences across both web and mobile environments.

5. MODULES OF TRAINING

Module	Learning / Task Completion	Tools / Technologies Used
Orientation (21st May)	Internship induction, orientation, introduction to company workflow	–
Development Setup (22nd May)	Installed and configured development environment	Node JS, VS Code, MYSQL, Git
Web Fundamentals (23rd – 27th May)	Learned HTML, CSS, React.js basics, Next.js routing, Tailwind CSS for styling	HTML, CSS, React.js, Next.js, Tailwind CSS
Frontend Development (28th May – 3rd June)	Built login & registration pages, dashboard UI, form validation, API integration	React.js, Next.js, Tailwind CSS, Axios/Fetch
Backend Development (4th – 10th June)	Created RESTful APIs, connected to MySQL, performed CRUD operations, integrated frontend with backend	Node.js, Express.js, MySQL, Postman
User Management System (10th June)	Completed login/registration system with database connectivity	Node.js, MySQL, React.js
Mobile App Development (11th – 15th June)	Created Flutter mock app with login, dashboard, and navigation	Flutter, Dart
Final Deployment (16th – 18th June)	Debugged, deployed project on Vercel, tested database connectivity, prepared documentation	Vercel, MySQL, Node.js
Project Review & Completion (19th – 21st June)	Reviewed project with mentor, submitted final report, received internship certificate	–

6. PROJECT WORK

Task 1: Full-Stack Web Application Development

Description:

As part of this task, I built a ZACRM-based order management system. The application allowed users to create new sales orders, store them in the database, and manage them through an interactive UI.

- The frontend was developed using Next.js and React.js**,** styled with Tailwind CSS for a professional and responsive design.
- The backend was implemented in Node.js with Express.js, where RESTful APIs were created to handle sales order operations.
- The MySQL database was used to store order details, customer data, and status updates.
- Sequelize ORM was integrated to manage schema design, database queries, and model synchronization.
- Data was synced between the ZACRM database and the ZAPPI database, ensuring smooth order flow between systems.

Features Implemented:

- **Add Sales Order** – Form to add customer details, order number, amount, and timestamps.
- **Order Management** – Dashboard to view all orders with filters, status updates, and action buttons.
- **Backend APIs** – Developed endpoints for creating, updating, retrieving, and deleting orders.
- **Database Integration** – Orders were stored in ZACRM database and synced with ZAPPI database**.**

- **Responsive UI** – Designed using Tailwind CSS to ensure accessibility across devices.

Tools and Technologies Used:

- **Frontend**: Next.js, Tailwind CSS
- **Backend**: Node.js, Express.js
- **Database**: MySQL with Sequelize ORM

Task 2: Flutter Mock Application Development

Description:

As part of the second project during my internship, I created a Flutter-based mock app. The app was designed as a prototype to demonstrate how a chatbot agent could interact with users in a mobile environment.

The mock app displayed a conversation interface, where a pre-programmed chatbot (e.g., fruit_orchard_agent) welcomed the user and provided assistance. The design included a chat-style UI with message bubbles, input fields, and send options.

This project helped me understand the Flutter framework**,** Dart programming language**,** and the structure of mobile applications including widgets, state management, and navigation.

Features Implemented:

- **Chat Screen UI** – Created a conversational UI with message bubbles and timestamps.
- **Predefined Responses** – The mock agent responded with pre-set messages such as *"*Hello! How can I assist you today*?"*.
- **User Input** – Users could type or send messages through the input box.

- **Agent Identity** – The chatbot was displayed with a name (e.g., fruit_orchard_agent) at the top for context.
- **Prototype Purpose** – Focused on UI/UX demonstration rather than backend API integration.

Tools and Technologies Used:

- **Framework**: Flutter
- **Language**: Dart
- **Platform**: Android (Tested on mobile device)

7. CONCLUSION

During my internship at Incisive Solutions, I successfully completed two projects that enhanced both my technical and professional skills. The first project was a full-stack web application, ZACRM Order Management System**,** developed using Next.js, React.js, Tailwind CSS, Node.js, Express.js, MySQL, and Sequelize ORM, where I implemented sales order management, database synchronization, and API integration.

The second project was a Flutter-based chatbot mock application, which strengthened my understanding of mobile app UI/UX design and cross-platform development. Alongside technical expertise, I also improved essential soft skills such as problem-solving, time management, communication, adaptability, and teamwork.

Overall, this internship was a valuable learning experience that bridged the gap between academic knowledge and industry practices, preparing me to take on future challenges in full-stack and mobile application development with confidence.

8. APPENDICES

A. CODING

```
import React, { useState, useEffect } from 'react';

import toast from 'react-hot-toast';

import { CalendarDays } from 'lucide-react';

import { withSessionSsr } from '../../lib/withSession';

import { useRef } from 'react';

export default function OrderManagement({ orders: initialOrders = [], date: initialDate =
'', syncedOrderId = '' }) {

  const [search, setSearch] = useState('');

  const [orders, setOrders] = useState(initialOrders);

  const [date, setDate] = useState(initialDate);

  const [selectedOrders, setSelectedOrders] = useState([]);

  const [selectAll, setSelectAll] = useState(false);

  const [displayLoading, setDisplayLoading] = useState(false);

  const [showPopup, setShowPopup] = useState(false);

  const [selectedOrderId, setSelectedOrderId] = useState(null);

  const [syncing, setSyncing] = useState(false);

  const hasShownToast = useRef(false);

  useEffect(() => {

    setOrders(

      search
```

```
        ? initialOrders.filter(o =>
            (o.increment_id || o.zaapko_order_id || '')
              .toLowerCase()
              .includes(search.toLowerCase())
          )
        : initialOrders
    );
  }, [search, initialOrders]);
useEffect(() => {
  if (syncedOrderId && !hasShownToast.current) {
    hasShownToast.current = true;
    toast.success(`Order ${syncedOrderId} synced successfully`);
    const url = new URL(window.location.href);
    url.searchParams.delete('syncOrderId');
    window.history.replaceState({}, '', url.toString());
  }
}, [syncedOrderId]);
  const handleDisplay = () => {
    setDisplayLoading(true);
    setTimeout(() => window.location.href = `/ordermanagement?date=${date}`, 1000);
  };
  const handleSelectAll = () => {
```

```
    if (selectAll) {
      setSelectedOrders([]);
    } else {
      setSelectedOrders(orders.map(o => o.increment_id || o.zaapko_order_id));
    }
    setSelectAll(!selectAll);
  };
  const handleCheckboxChange = (orderId) => {
    setSelectedOrders(prev =>
      prev.includes(orderId) ? prev.filter(id => id !== orderId) : [...prev, orderId]
    );
  };
  const handleSyncOrder = (orderId) => {
    window.location.href = `/ordermanagement?date=${date}&syncOrderId=${orderId}`;
  };
  return (
    <div className="min-h-screen bg-[#f9f9f9] p-6 text-gray-900">
      <div className="flex flex-col sm:flex-row sm:justify-between gap-4 mb-6">
        <input
          type="text"
          placeholder="Search Order ID"
          className="border border-gray-300 rounded px-4 py-2 sm:w-64 focus:outline-
none"
```

```
          value={search}
          onChange={(e) => setSearch(e.target.value)}
        />
        <div className="flex gap-2 items-center">
          <button
            onClick={handleDisplay}
            disabled={displayLoading}
      className="bg-green-600 text-white rounded px-4 py-2 hover:bg-green-700 transition>
              {syncing ? 'Syncing...' : 'Yes'}
            </button>
          </div>
      )}
    </div>
  );
}


export const getServerSideProps = withSessionSsr(async ({ query }) => {
  const date = query.date || new Date().toISOString().slice(0, 10);
  const syncOrderId = query.syncOrderId || null;


  try {
    console.log("Entering Props....");
```

```
  const res = await fetch(`${process.env.NEXTAUTH_URL}/api/orders`, {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
    },
    body: JSON.stringify({ specific_date: date, syncOrderId }),
  });
  const result = await res.json();
  const orders = result.orders || [];
  return {
    props: {
      orders,
      date,
      syncedOrderId: syncOrderId || '',
    },
  };
} catch (err) {
  console.error('SSR Fetch Error:', err);
  return { props: { orders: [], date, syncedOrderId: '' } };}
```

B. SCREENSHOTS

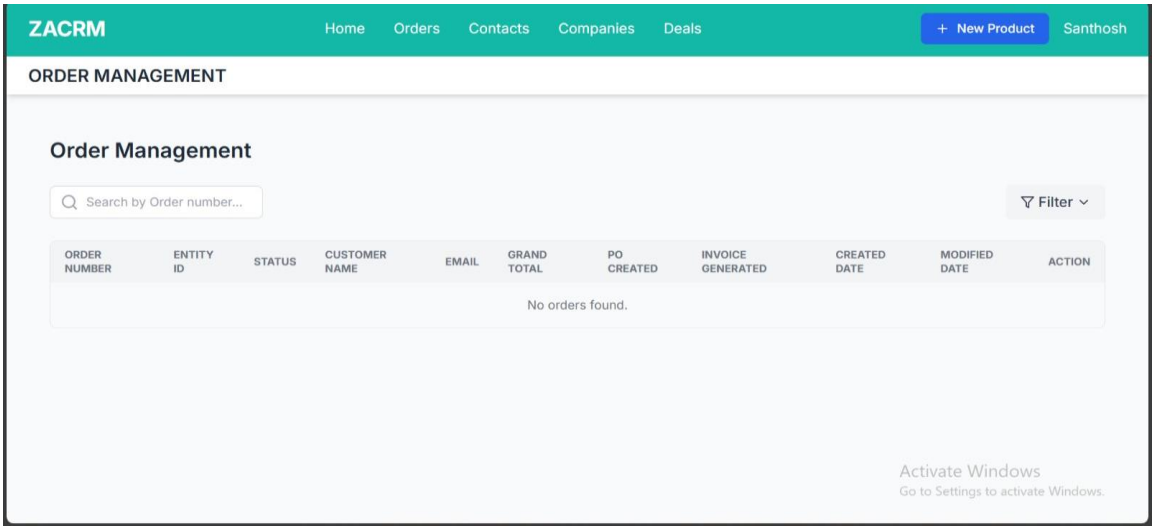


FIG : Zaapi Orders Page

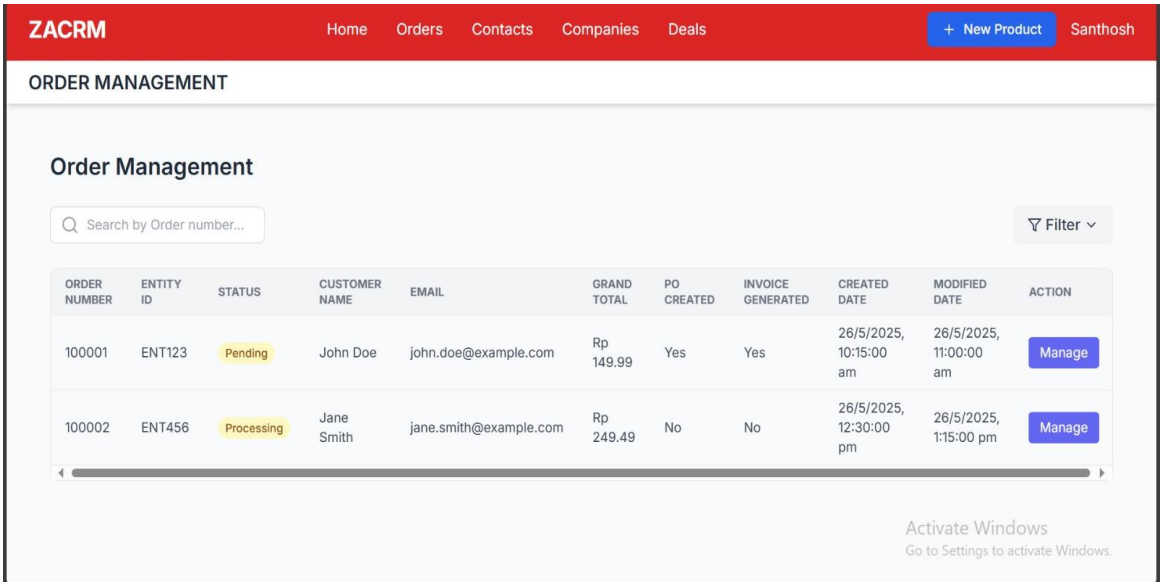


FIG : Zacrm Orders Page

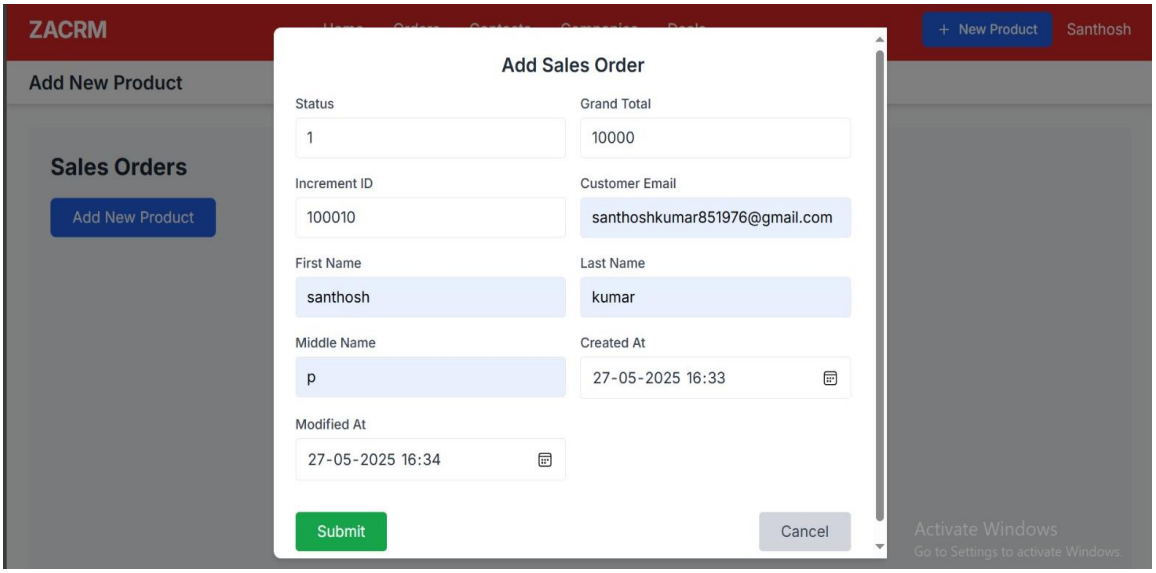


FIG : Zacrm Sales Order Page

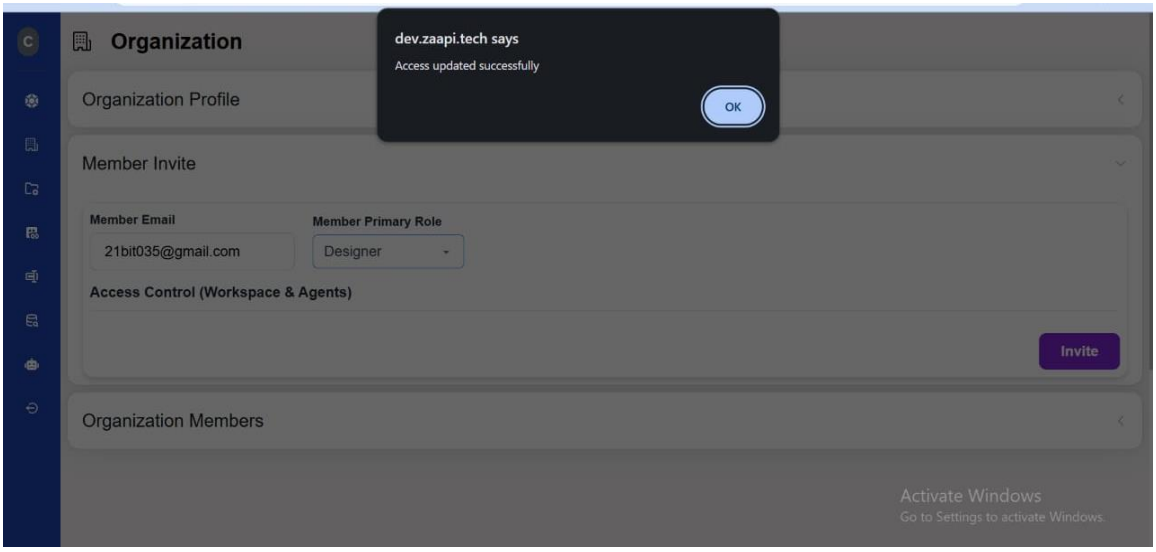


FIG : Testing & Training

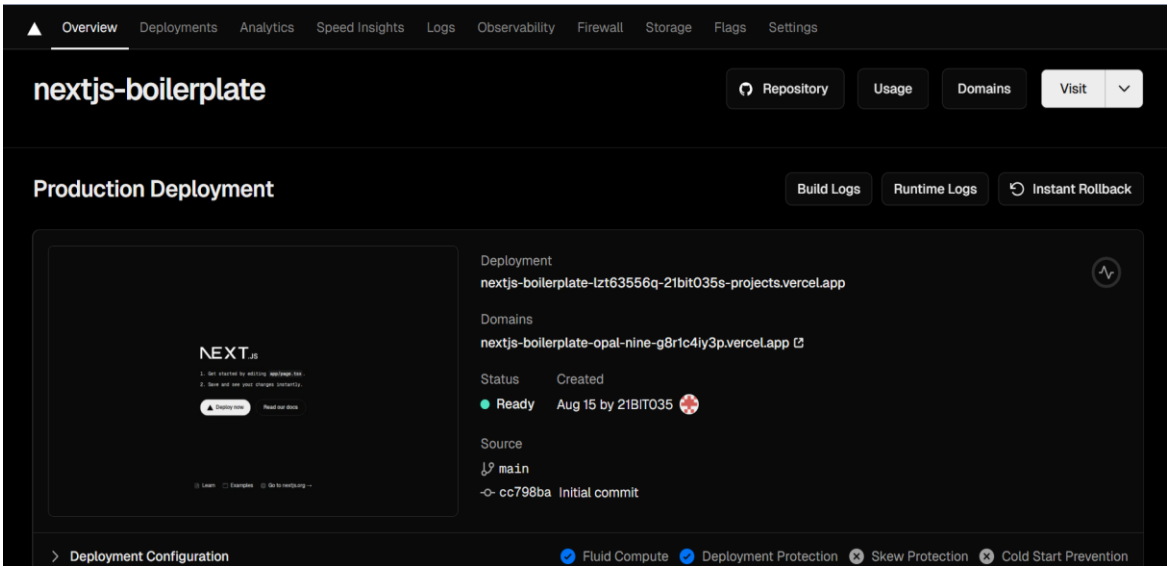
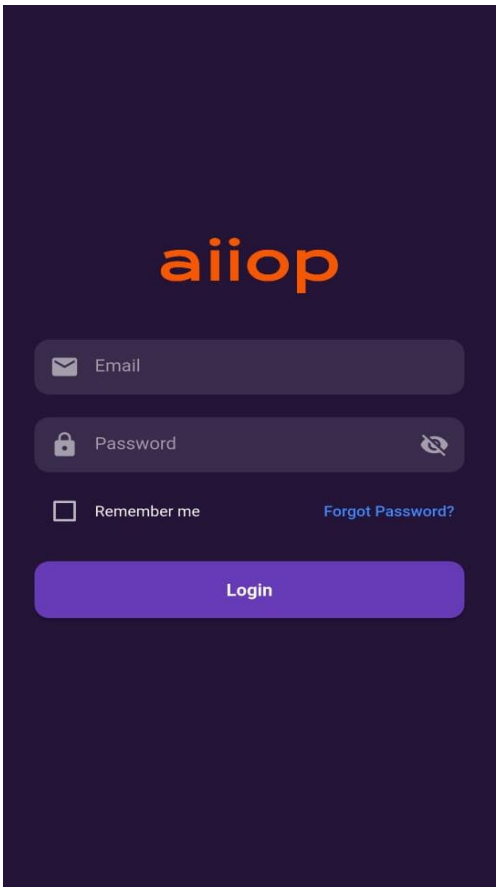


FIG : Deployment in Vercel



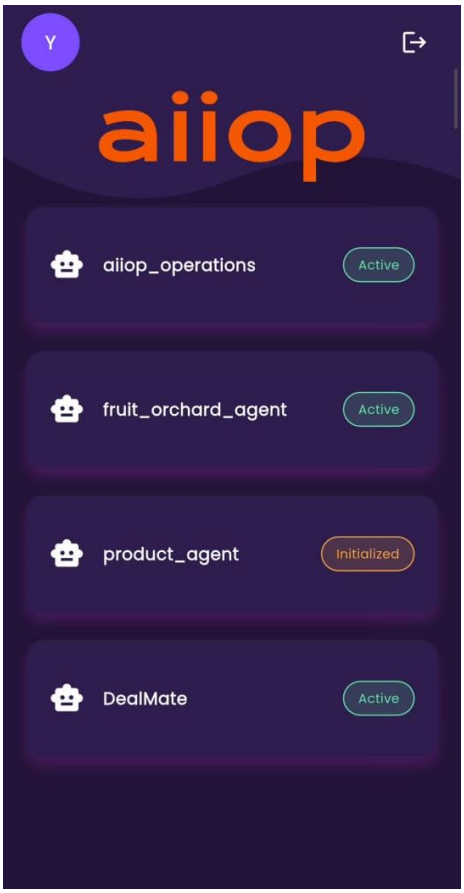
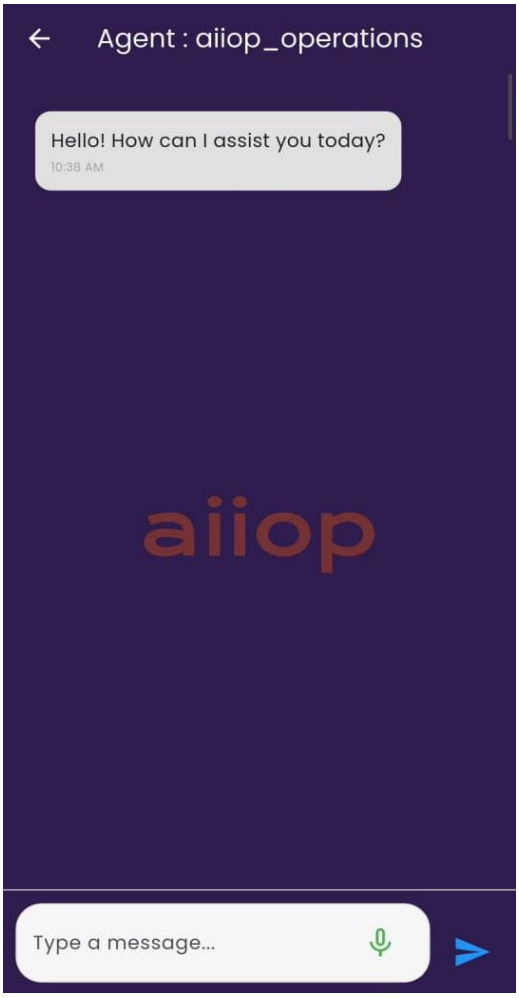


FIG : aiiop Login & Dashboard



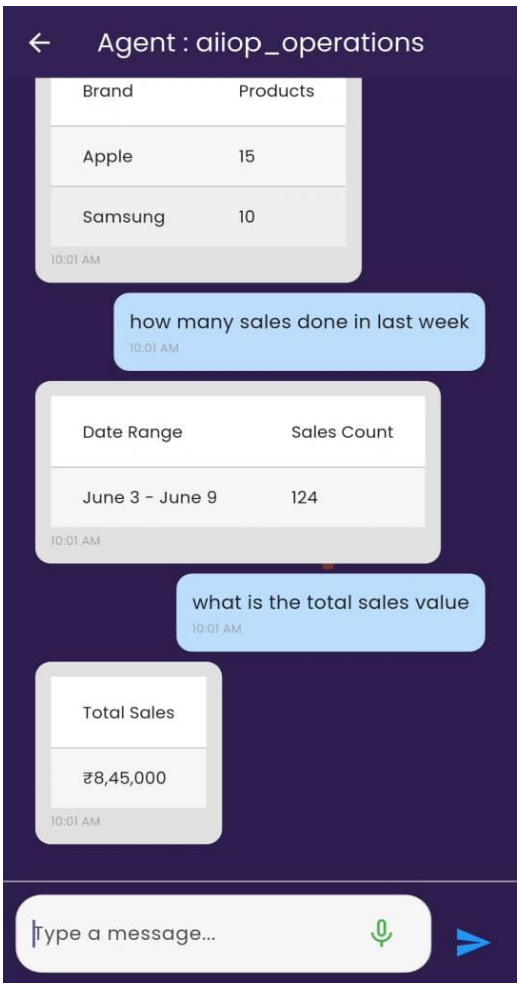


FIG : aiiop Chat Application in English

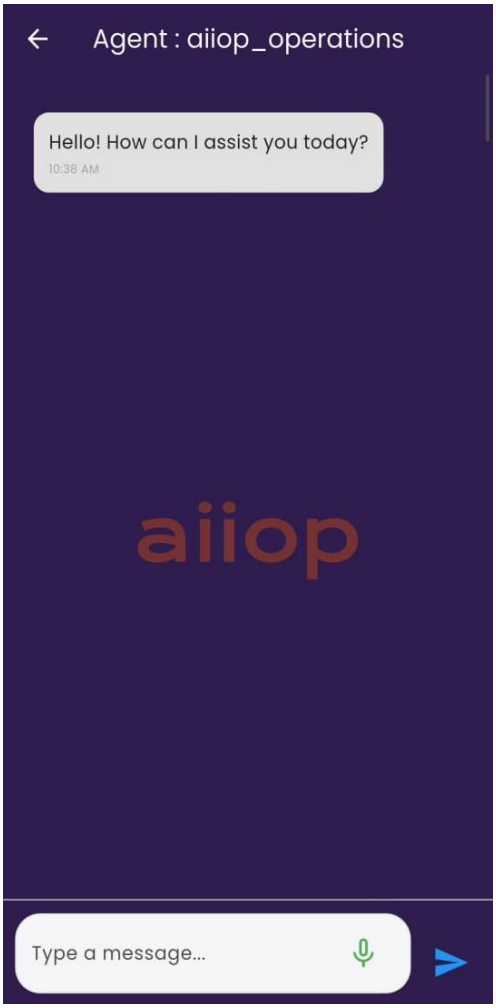


FIG : aiiop Chat Application in Tamil